

TIANRUN ZHAO

New Computer Science Building, Stony Brook University, Stony Brook, NY 11794

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Education

Stony Brook University, *Dept. of Computer Science, Stony Brook, NY*

2024 – present

Nanjing University, *School of Physics, Nanjing, China*

2020 – 2024

- Major GPA: **3.73** / **4.0** Ranking: **3** / **144**
- *Awards & Scholarships*: First Prize in China Undergraduate Physics Experiment Competition; Scholarship of Elite Program of Basic Sciences

Publications

Formalizing CHSH Rigidity in Lean 4

Tianrun Zhao et al. *arXiv preprint* arXiv:2604.03884 April 2026

Formalized in Lean 4 a rigidity theorem showing that any strategy achieving near-optimal CHSH value must be locally isometric to the canonical qubit strategy; identified and addressed a gap in the argument of McKague, Yang, and Scarani. arxiv.org/abs/2604.03884

Industry Experience

Software Engineering Intern, *Cadence Design Systems, Austin, TX*

Summer 2025

- Deployed and evaluated a MuJoCo-based physics simulation environment for a car model; benchmarked simulation speed and physical accuracy against alternative simulation frameworks.
- Extended the simulation pipeline to model deformable objects, implementing and testing a rubber band simulation within the MuJoCo environment.
- Integrated the full workflow into a small-scale autonomous car simulation, validating vehicle dynamics in a controlled virtual testbed.

Research Experience

Entanglement Features Generalization, *UCSD (Remote)*

Mar. 2023 – Jan. 2024

Supervised by Prof. Yi-Zhuang You

- Generalized entanglement features to the Lindblad equation; used vectorization to transform a superoperator into an operator form.
- Calculated the Ginibre ensemble average of a Lindblad gate using random matrix theory, capturing the random dynamics of a Lindblad evolution.
- Computed the entanglement feature of a simple Lindblad propagator, characterizing entanglement properties of open quantum systems.

- Applied the Gauss–Bonnet theorem from differential geometry to Berry's phase verification; compared classical and non-classical theories and validated results experimentally.
- Designed an optical coil sensor using a large spring and optical fibers capable of detecting mechanical stress and water depth with high accuracy.
- Built a fiber model and estimated experimental errors; verified sensor stability through MATLAB simulations.